

What is DevOps?

DevOps is a combination of Development (Dev) and Operations (Ops). It is a software development approach that helps developers and IT operations teams work together to build, test, release, and maintain applications faster and more reliably.

The main goal of DevOps is to deliver high-quality software quickly with fewer errors by using automation, continuous integration, continuous delivery, and continuous monitoring.

Definition

DevOps is a culture, methodology, and set of practices that integrates software development and IT operations to improve collaboration, automate workflows, and deliver software rapidly and reliably.

Objectives of DevOps

- Improve collaboration between development and operations teams.
- Automate software development and deployment processes.
- Deliver software faster.
- Reduce deployment failures.
- Improve software quality.
- Increase customer satisfaction.
- Enable continuous improvement.

Benefits of DevOps

- Faster software delivery
- Better collaboration
- Improved software quality
- Reduced deployment time
- Early bug detection

- Higher customer satisfaction
- Better security through automation
- Faster recovery from failures
- Increased productivity
- Reduced operational costs

DevOps Lifecycle

The DevOps lifecycle is a continuous process consisting of several stages.

Plan



Develop



Build



Test



Release



Deploy



Operate



Monitor



(Repeat)

1. Plan

Purpose

The project requirements are gathered, analyzed, and planned.

Activities

- Requirement gathering
- Project planning
- Sprint planning
- Task assignment
- Risk analysis

Tools

- Jira
- Azure Boards
- Trello

2. Develop

Purpose

Developers write the application code.

Activities

- Coding
- Code review
- Version control

Tools

- Visual Studio
- VS Code
- Eclipse
- Git

- GitHub

3. Build

Purpose

Convert source code into executable software.

Activities

- Compile source code
- Resolve dependencies
- Package application

Tools

- Maven
- Gradle
- MSBuild
- Ant

4. Test

Purpose

Automatically test the software to identify bugs before deployment.

Types of Testing

- Unit Testing
- Integration Testing
- Functional Testing
- Performance Testing
- Security Testing

Tools

- Selenium

- JUnit
- NUnit
- TestNG

5. Release

Purpose

Prepare the application for deployment.

Activities

- Version management
- Approval process
- Release planning

Tools

- Jenkins
- GitHub Actions
- Azure DevOps

6. Deploy

Purpose

Deploy the application to production or testing environments.

Activities

- Automated deployment
- Configuration management
- Rollback if deployment fails

Tools

- Docker
- Kubernetes

- Ansible
- Terraform

7. Operate

Purpose

Run and maintain the application in production.

Activities

- Server management
- Database management
- Backup
- Performance tuning

Tools

- Kubernetes
- Docker
- Linux

8. Monitor

Purpose

Monitor application performance and infrastructure health.

Activities

- Performance monitoring
- Error tracking
- Log analysis
- Alerting

Tools

- Prometheus

- Grafana
- ELK Stack
- Splunk

Popular DevOps Tools

Stage	Tools
Planning	Jira, Trello, Azure Boards
Coding	Git, GitHub, Bitbucket
Build	Maven, Gradle, MSBuild
Testing	Selenium, JUnit, NUnit
CI/CD	Jenkins, GitHub Actions, GitLab CI, Azure DevOps
Containerization	Docker
Orchestration	Kubernetes
Configuration	Ansible, Puppet, Chef
Monitoring	Prometheus, Grafana, ELK

What is CI/CD?

CI/CD stands for:

- **CI** – Continuous Integration
- **CD** – Continuous Delivery or Continuous Deployment

CI/CD is a DevOps practice that automates the process of integrating code changes, testing them, and deploying them to different environments.

It helps developers deliver software faster, more frequently, and with fewer errors.

Continuous Integration (CI)

Definition

Continuous Integration is the practice of frequently merging code changes into a shared repository, where automated builds and tests verify the changes.

CI Process

Developer Writes Code



Push Code to Git Repository



CI Tool Detects Changes



Build Application



Run Automated Tests



If Successful → Merge Code

Benefits of CI

- Detects bugs early
- Improves code quality
- Reduces integration issues
- Enables frequent code commits
- Saves development time

Popular CI Tools

- Jenkins
- GitHub Actions
- GitLab CI
- Azure DevOps

- CircleCI
- Travis CI

Continuous Delivery (CD)

Definition

Continuous Delivery automatically prepares tested code for release. The software is always in a deployable state, but deployment to production usually requires manual approval.

Continuous Delivery Flow

Code



Build



Test



Package



Manual Approval



Deploy to Production

Benefits

- Faster releases
- Lower deployment risk
- Reliable software delivery
- Easy rollback

Continuous Deployment

Definition

Continuous Deployment goes one step further by automatically deploying every successful change to production without manual approval.

Continuous Deployment Flow

Code



Build



Test



Package



Automatic Deployment



Production

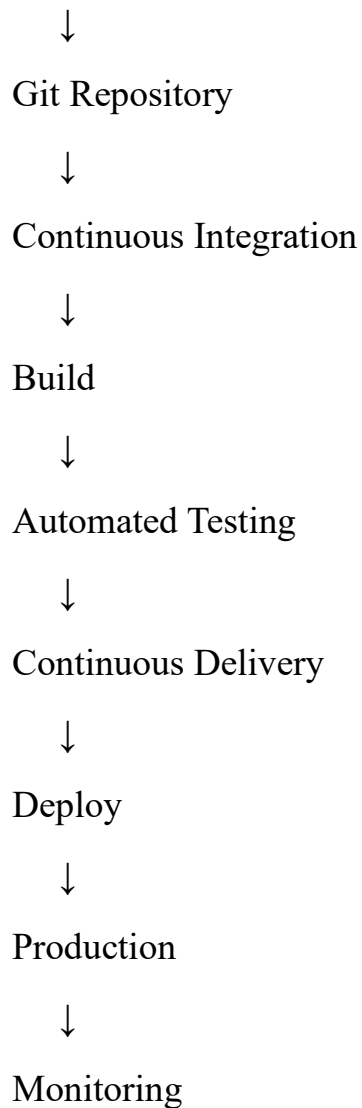
Difference Between Continuous Delivery and Continuous Deployment

Feature	Continuous Delivery	Continuous Deployment
Manual approval	Required	Not required
Deployment	Manual after approval	Automatic
Risk	Lower	Higher if testing is weak
Speed	Fast	Very fast

CI/CD Pipeline

A typical CI/CD pipeline consists of these stages:

Developer



Popular CI/CD Tools

Tool	Purpose
Jenkins	Open-source automation server for CI/CD
GitHub Actions	CI/CD workflows integrated with GitHub
GitLab CI/CD	Built-in CI/CD for GitLab projects
Azure DevOps	End-to-end DevOps platform with CI/CD pipelines
CircleCI	Cloud-based CI/CD automation
Bamboo	CI/CD tool by Atlassian

DevOps vs CI/CD

DevOps

A culture and methodology

Covers the complete software lifecycle

Includes collaboration, automation, monitoring, and operations

Involves people, processes, and tools

CI/CD

A practice within DevOps

Focuses on build, test, and deployment automation

Focuses on integrating and delivering code continuously

Mainly involves automation tools and pipelines